M.SUBHI ABO RDAN

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Education

Massachusetts Institute of Technology (MIT) Cambridge, MA

Master of Engineering in Electrical Engineering and Computer Science *June 2027*

Bachelor of Science in Computer Science and Engineering *June 2026*

Bachelor of Science in Physics *June 2026*

- **EECS Coursework:**
  Digital Systems Laboratory (FPGA), Accelerated Computing (GPUs), Software Performance Engineering, Computer Architecture, Distributed Systems, Operating Systems.
- **Theory and Physics Coursework:** Quantum Computation, Quantum Information Science II, Quantum Engineering Platforms, Quantum Mechanics, General Relativity, Statistical Mechanics, Complexity and Computability Theory, Statistical Data Analysis.

Research and Work Experience

Optiver June 2026 – August 2026

FPGA Engineering Intern *Chicago, IL*

- Incoming High Frequency Trading FPGA Engineering Intern for the summer of 2026.

Apple May 2025 – August 2025

Silicon Prototyping Engineering Intern — FPGA Engineer *Cupertino, CA*

- Built a non-invasive distributed clock monitoring system for multi-board FPGA setups, enabling run-time frequency measurement and verification of external clocks via a register-mapped UMRBus3 interface.
- Engineered a cycle-accurate FPGA validation engine using RTL waveform replays to emulate the rest of the system, enabling high-throughput DUT testing.
- Designed a software/hardware execution pipeline to parse and compress VCD transition traces (220× reduction) and drive FPGA-based DUT validation.

MIT Space Systems Laboratory — STAR Lab June 2024 – August 2025

Research Assistant — FPGA Engineer *Cambridge, MA*

- Designed, implemented, and performance-tested an optical modeling system for high-order wavefront sensing and control algorithms on the AMD Zynq UltraScale+ MPSoC ZCU104 using High-Level Synthesis (HLS).
- Contributed to a **first-author publication at SPIE Photonics West 2025**: "Accelerating High-Order Wavefront Sensing and Control Algorithms with FPGAs".
- Designed a 2D FFT accelerator, achieving a 360× improvement in execution time over the Arm Cortex A53 CPU.
- Developed an Angular Spectrum Method accelerator interfacing with the CPU via Xilinx Runtime (XRT) C++ API, optimizing dataflow, reducing data movement, and increasing parallelism, achieving a 500× speedup over CPU.

MIT Multifunctional Metamaterials — META Lab Feb 2024 – Jun 2024

Research Assistant *Cambridge, MA*

- Developed a physical model to investigate the snap-in dynamics of microcantilevers in Atomic Force Microscopes.
- Constructed a computational simulation utilizing finite difference method. This simulation predicts the optimal snap-in time and distance critical for achieving high-resolution nano-scale measurements in AFMs.

MIT Media Lab – Signal Kinetics Jun 2023 – Sep 2023

Research Assistant *Cambridge, MA*

- Designed a simulation using Python based on the Jaffe and McGlamery underwater imaging model to generate underwater images of the Macbeth ColorChecker that accurately replicates real-world underwater conditions.
- Developed an algorithm capable of reconstructing color images by integrating information from multiple monochromatic underwater images taken at different distances. On going evaluation with real data.

Projects

Intermediate-Code Automorphisms in Pieceable Fault-Tolerant Quantum Computation May 2026

- Developed a framework for studying dynamically evolving stabilizer-code symmetries during pieceable fault-tolerant quantum gate constructions, focusing on logical Clifford and multi-qubit gate synthesis.
- Implemented automorphism-search and logical-action verification tools using Python, Stim, and GF(4)/symplectic representations to analyze intermediate quantum error-correcting codes and encoded logical operations.

Near-Memory FPGA 100G-Ethernet UDP Engine December 2025

- Designed and implemented a low-latency, line-rate 100 Gbps UDP offload engine on an AMD UltraScale+ RFSoC operating at 322.22 MHz, including connection bookkeeping via a fully pipelined, set-associative hash table.
- Built a SmartNIC-style near-memory architecture leveraging AXI HP and cache-coherent ACP ports via DMA, enabling ultra-low-latency packet delivery directly into CPU memory and caches.

Fault-Tolerant Sharded Key-Value Store April 2025

- Implemented the Raft distributed-systems consensus algorithm. Integrated snapshotting to compress Raft logs, reducing recovery latency and minimizing storage footprint.
- Built a distributed, fault-tolerant key-value store with dynamic sharding using Raft, supporting reconfiguration and load balancing via RPC-based coordination among cluster nodes.

FPGA QR-Code Reader December 2023

- Developed a real-time QR code scanner on an AMD Urbana FPGA, integrating an OV7670 camera module and HDMI output; awarded the **MIT EECS George C. Newton Laboratory Prize** (2024).
- Implemented image processing pipelines to grayscale, denoise, detect alignment patterns, and decode QR codes as small as 240×240 pixels in under 0.02 seconds.

Publications

- **M. Subhi Abo Rdan**, N. Belsten, S. Rao, A. W. Taka, B. Eickert, K. Milani, E. S. Douglas, L. Pogorelyuk, K. Cahoy, "Accelerating high-order wavefront sensing and control algorithms with FPGAs," Proc. SPIE 13627, 2025. DOI: 10.1117/12.3064675
- B. Eickert, N. Belsten, K. Milani, A. Taka, **M. Subhi Abo Rdan**, L. Pogorelyuk, S. Rao, E. S. Douglas, K. Cahoy, "Demonstrating real-time wavefront sensing and control with space processors," Proc. SPIE 13627, 2025. DOI: 10.1117/12.3065650

Honors & Awards

- MIT EECS George C. Newton Laboratory Prize, 2024
- International Physics Olympiad (IPhO), Bronze Medalist, 2020 & 2021
- European Physics Olympiad (EuPhO), Bronze Medalist, 2020 & 2021
- International Applied Physics Olympiad (IAPhO), Silver Medalist, 2021
- Physics Cup, Honorable Mention, 2021

Skills and Interests

Languages: C/C++, SystemVerilog, Python, Go, CUDA, TypeScript, RISC-V Assembly, TCL.

Hardware & Systems: FPGA Design, High-Level Synthesis (HLS), Computer Architecture, Distributed Systems, Networking, SmartNICs, GPU Computing, Quantum Error Correction.

Tools/Frameworks: Vivado, Vitis, XRT, ProtoCompiler, Cocotb, Pynq, Stim, Git, Linux/UNIX.

Other Experience: FPGA Teaching Assistant, Quantum Physics Grader, Citadel Securities Quantitative Invitational.

Activities: Weight Lifting, Swimming, Ice Hockey, Tennis, Soccer.